

# The NFSC Pedagogy: Narrative-Driven Cognitive Progression in  
Graduate Engineering Education

## A Case Study from a Chinese Solid Waste Management Course

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**\*\*Abstract:\*\*** Engineering graduate education faces a well-documented challenge: the disconnection between decontextualised textbook cases and the multidimensional complexity of authentic engineering problems. This paper presents the NFSC (Narrative - Framework - Scenario - Creation) pedagogy, a four-dimensional progressive teaching model developed and iterated over three academic years (2023 - 2025) in a graduate solid waste management course at a Chinese university. Drawing on Bruner's narrative theory, constructivist learning principles, situated learning theory, and outcome-based education, NFSC structures the learning process into a cognitive progression: narrative excerpts from documentary novels trigger affective engagement; a domain-specific Mismatch - Repair analytical framework provides diagnostic tools; three types of immersive scenarios (EIA debate, Nudge design, mock environmental court) temper application capabilities; and a three-track output system closes the learning loop. Using a practitioner case study design with thematic analysis of student outputs and classroom observations, we document three forms of

cognitive change: shifts in causal attribution from intuitive moral judgment to framework-informed analysis, longitudinal growth in multi-dimensional framework usage, and diversification of creative outputs. The study's limitations—single course, small cohorts (~30 per cycle), absence of controlled comparison—are honestly acknowledged. We discuss transferability preconditions for other engineering disciplines and position this work as exploratory practitioner research that may inform larger-scale investigation.

**\*\*Keywords:\*\*** engineering education; graduate teaching; narrative pedagogy; scenario-based learning; constructive alignment; practitioner research

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## ## Introduction

Engineering graduate education worldwide faces a structural tension. Accreditation frameworks increasingly demand competency-based outcomes—what graduates can *do*, not merely what they *know* (Crawley et al., 2014; Spady, 1994). Yet the dominant instructional paradigm remains the lecture, which excels at efficient knowledge transmission but struggles to cultivate the multidimensional judgment required in authentic engineering practice (Biggs & Tang, 2011; Laurillard, 2012).

This tension manifests acutely in cases used for teaching. Textbook cases are typically “clean problems”: abstracted, decontextualised, and stripped of the political, social, and temporal complexity that characterises real engineering work. By contrast, authentic engineering practice involves “wicked problems” (Lönngren & van Poeck, 2021): a landfill siting decision, for instance, involves not only geological and hydrological analysis but also community opposition, trans-jurisdictional accountability ambiguity, and temporal delays in

environmental benefit realisation.

Prior pedagogical frameworks have addressed aspects of this challenge. Problem-based learning (PBL) situates learning within authentic problems (Hmelo-Silver, 2004); the CDIO framework emphasises conceiving, designing, implementing, and operating real-world systems (Crawley et al., 2014); case-based teaching provides contextualised analytical practice (Herreid, 2011). However, these approaches typically assume that students already possess the analytical frameworks needed to diagnose presented problems, and that the motivational dimension—why students *\*care\** about the problem—is either self-evident or incidental.

The NFSC (Narrative - Framework - Scenario - Creation) pedagogy addresses this gap by explicitly building the motivational foundation *\*before\** introducing analytical tools, and by structuring the entire learning experience as a cognitive progression rather than a collection of activities. Developed in a graduate course “Solid Waste Pollution Prevention and Control Technology” at Chengdu University, China, NFSC draws upon seven documentary network novels—non-fiction prose totalling approximately two million characters—published by the course instructor as a narrative resource pool. Twenty-one teaching excerpts (coded S001–S021) selected from these novels provide authentic, emotionally textured accounts of real solid waste management conflicts across China.

This paper presents the theoretical grounding, instructional design, and eight-week implementation of NFSC, documenting cognitive changes observed across three teaching cycles (2023–2025). The study adopts a practitioner case study approach (Yin, 2018), using qualitative analysis of student outputs and classroom observations to provide “fieldnote” evidence of pedagogical effects. We make no causal claims—the inquiry is exploratory, positioning NFSC as a teaching innovation worthy of broader investigation rather than a proven intervention.

Our research questions are: (1) How does the NFSC progression influence students' causal attribution patterns when diagnosing solid waste management problems? (2) To what extent does framework usage develop longitudinally across the eight-week cycle? (3) What forms of creative output emerge from a three-track assessment system?

## ## Literature Review

### ### Engineering Education Reform and the Competency Gap

Engineering education has undergone substantial reform since the turn of the millennium, driven by accreditation standards that foreground competency-based outcomes. The Washington Accord (International Engineering Alliance, 2014) and subsequent national adaptations have institutionalised the expectation that graduates demonstrate not only technical knowledge but also design capability, ethical reasoning, and systems thinking. The CDIO initiative (Crawley et al., 2014) operationalised these expectations through a structured framework emphasising conceiving, designing, implementing, and operating—a significant advance over traditional lecture-based curricula. Yet CDIO and similar frameworks are primarily *\*curriculum-level\** interventions; they provide less guidance on *\*course-level\** pedagogical strategies for individual instructors seeking to transform their own classrooms.

At the course level, the “last mile” problem persists: how to translate abstract competency statements into concrete, week-by-week instructional sequences (Biggs & Tang, 2011). Constructive alignment theory (Biggs, 1996) articulates the principle—learning activities and assessments must align with intended learning outcomes—but leaves the specific pedagogical *\*mechanisms\** underspecified. How, precisely, does an instructor design activities that cultivate diagnostic thinking, systems reasoning, and ethical judgment within the constraints of a standard semester?

### ### Narrative Pedagogy in Higher Education

Narrative theory offers one answer. Bruner (1986) distinguished between paradigmatic thinking (logical, propositional) and narrative thinking (story-based, contextual), arguing that humans organise experience and construct meaning primarily through narrative structures. In educational contexts, narrative pedagogy has been shown to enhance engagement, deepen comprehension, and facilitate the integration of abstract concepts with lived experience (Goodson & Gill, 2011; McDrury & Alterio, 2003).

In engineering education specifically, the potential of narrative remains underexplored. While case-based teaching is well-established (Herreid, 2011; Yin, 2018), cases are typically presented as \*analytical objects\*—"read the background, answer the questions"—rather than as \*narrative experiences\* that engage students affectively before engaging them analytically. This ordering matters: cognitive science research suggests that emotional engagement enhances memory encoding and retrieval (Immordino-Yang & Damasio, 2007), implying that pedagogical designs that trigger emotional resonance \*before\* introducing analytical frameworks may enhance learning outcomes.

### ### Scenario-Based Learning and Immersive Practice

Situated learning theory (Lave & Wenger, 1991) posits that knowledge acquisition cannot be divorced from its application context. This principle underlies scenario-based learning approaches, which place students in authentic professional situations requiring decision-making under constraints (Errington, 2011). In engineering education, scenarios have been employed in various forms—role-plays, simulations, design challenges—with documented benefits for professional skill development (Bédard et al., 2012).

However, scenario-based learning typically functions as a standalone activity rather than as one dimension of an integrated pedagogical progression. Students may participate in a simulation or role-play, but the activity is rarely systematically connected to prior narrative engagement or subsequent creative production. The NFSC model addresses this gap by positioning scenario-based learning as the \*third\* dimension of a four-stage progression, explicitly building on narrative-triggered motivation and framework-equipped analytical capability.

### ### Practitioner Research in Engineering Education

This study adopts a practitioner case study approach (Cochran-Smith & Lytle, 2009; Yin, 2018), wherein the instructor-researcher systematically documents and analyses teaching practice. Practitioner research occupies a distinct epistemological position: it does not claim generalisability through statistical inference, but rather offers "thick description" (Geertz, 1973) of situated practice that may inform other practitioners' reflection and adaptation. This approach has gained legitimacy in engineering education research (Borrego & Bernhard, 2011), particularly for exploratory studies of novel pedagogical innovations where controlled experimental designs are premature or infeasible.

## ## Methodology

### ### Research Design

This study employs a qualitative practitioner case study design (Yin, 2018) with thematic analysis of student outputs and classroom observations (Braun & Clarke, 2006). The research is positioned as exploratory, documenting a pedagogical innovation across three

iterative teaching cycles without making causal claims.

### ### Setting and Participants

The study was conducted in the graduate elective "Solid Waste Pollution Prevention and Control Technology" (course code 026039003, 2 credits, 32 contact hours) at Chengdu University, China, across three academic years (2023–2025, Autumn semesters). Each cycle enrolled approximately 30 master's students, with approximately 60% from environmental engineering backgrounds and 40% from cross-disciplinary backgrounds (chemistry, materials science, ecology). The course underwent three iterative revisions: v1.0 (2023, traditional lecture), v2.0 (2024, pilot NFSC), and v3.0 (2025, full NFSC implementation). Data reported in this paper draw primarily from the v2.0 and v3.0 cycles.

### ### The NFSC Pedagogy

The NFSC pedagogy structures the eight-week course into four progressive dimensions:

*\*Narrative (Week 1).\** Students pre-read excerpts from documentary network novels depicting authentic solid waste conflicts—landfill-induced land devaluation, technology-centrism in environmental impact assessments, rural waste classification failures, and community dignity claims in NIMBY disputes (Huang, 2025a). In-class discussion centres on affective engagement: "What did the villagers lose?" Analytical frameworks are deliberately withheld.

*\*Framework (Week 2).\** The instructor introduces the Mismatch–Repair framework, comprising six sub-frameworks (spatial mismatch, siting justice, technological mismatch, waste time, behavioural mismatch, humanistic mismatch) totalling 34 diagnostic dimensions (Huang,

2025b). Students re-analyse Week 1's narrative excerpts using the frameworks and compare their intuitive and framework-guided analyses.

**\*Scenario (Weeks 3 - 7).** Three immersive scenarios are deployed: an environmental impact assessment debate (Week 3, role-playing as technical experts, regulators, and community representatives), a Nudge-based behavioural intervention design (Weeks 5 - 6, designing and simulating policy instruments for rural waste classification), and a mock environmental court hearing (Week 7, litigating procedural versus substantive justice in facility siting). Role cards, randomised event cards, and instructor facilitation guides structure each scenario.

**\*Creation (Week 8).** Students apply the Five-Dimensional Unified Diagnostic Matrix (Space  $\times$  Technology  $\times$  Behaviour  $\times$  Humanism  $\times$  Time) to a comprehensive case and submit output through one of three tracks: Track A (research paper analysing a solid waste problem and discussing framework limitations), Track B (engineering design for a treatment facility), or Track C (pedagogical reflection on personal learning experience).

### ### Data Collection and Analysis

Data sources comprised: (a) classroom observation notes recorded by the instructor immediately after each session; (b) student written outputs including in-class analysis worksheets (Week 1 - 2), scenario reflection memos (Weeks 3 - 7), and final track outputs (Week 8); and (c) a zero-configuration learning portal that tracked voluntary student interactions with pre-class materials and AI-assisted learning prompts.

Data were analysed using thematic analysis (Braun & Clarke, 2006) following a six-phase process: familiarisation, initial coding, theme development, theme review, theme definition, and reporting. Two researchers independently coded a subset of student outputs; inter-



coder agreement exceeded 85% for all identified themes. Member checking was conducted with a subset of consenting students from the v3.0 cohort.

### ### Ethical Considerations

Institutional ethics approval was obtained. Students were informed that course outputs might be used for research purposes with anonymisation; opt-out was available without academic consequence. All names and identifying details have been removed.

## ## Findings

Three themes emerged from the analysis, corresponding to our three research questions.

### ### Theme 1: From Moral Attribution to Framework-Informed Diagnosis

A notable shift occurred in students' causal attribution patterns between Week 5 and Week 8. During Week 5's behavioural mismatch instruction, students were asked to intuitively diagnose why a rural waste classification programme failed despite adequate infrastructure. Among N = 28 respondents, the most frequent attribution was "low villager quality" (n = 11, 39%), followed by "insufficient publicity" (n = 6, 21%), and "habit change is difficult" (n = 5, 18%). The instructor introduced the Behavioural Mismatch Four-Factor framework (cognitive barrier, convenience trap, norm deficit, feedback delay) without commenting on students' initial attributions.

By Week 8, when a structurally similar problem was presented in a different context, only n = 2 (7%) invoked "low quality" attribution, while n = 7 (25%) cited "convenience trap," n = 6 (21%) cited

"feedback delay," and  $n = 4$  (14%) cited "norm deficit." One student wrote in their Week 8 reflection: "I used to think the classification problem was about villager awareness. Now I see the waste bins are placed 300 metres from the village entrance—the 'convenience trap' dimension makes the structural design failure visible."

This shift from intuitive moral judgment ("they don't care enough") to framework-informed structural analysis ("the system makes it hard for them to care") was observed across all three teaching cycles, with the v3.0 cohort showing the most pronounced change—attributable, we suggest, to the improved scenario design in later iterations.

### Theme 2: Progressive Framework Integration

Longitudinal assessment of framework usage was conducted at three time points across the eight-week cycle (Table 1). At Week 1 baseline, following initial exposure to the spatial mismatch framework, approximately half of students ( $n \approx 15$ ) were able to accurately use framework sub-dimension terminology in their written analyses. By Week 4, following exposure to four frameworks, approximately 25 students employed at least two framework dimensions when analysing a multi-dimensional case. By Week 8, approximately 12 students spontaneously integrated three or more dimensions for cross-dimensional interaction analysis—a pattern that did not appear in earlier teaching cycles using traditional lecture methods.

\*\*Table 1. Pre-Mid-Post Framework Usage Assessment\*\*

| Time Point    | N (approx.) | Key Metric                              | Students Meeting Criterion |
|---------------|-------------|-----------------------------------------|----------------------------|
| Week 1 (Pre)  | 30          | Accurate framework terminology          | ~15                        |
| Week 4 (Mid)  | 30          | $\geq 2$ framework dimensions           | ~25                        |
| Week 8 (Post) | 30          | $\geq 3$ dimensions with cross-analysis | ~12                        |

### Theme 3: Diversification of Creative Outputs

Across three teaching cycles, final outputs were distributed approximately as follows: Track A (research paper) 40%, Track B (engineering design) 35%, and Track C (pedagogical reflection) 25%. The Track C output—in which students analysed their own eight-week learning trajectory—proved particularly illuminating. Approximately 20% of students across all tracks demonstrated emerging metacognitive awareness in their reflections. One student in the Track C group articulated: “Before this course, I assumed solid waste problems were essentially technical. Now I recognise them as simultaneously spatial, institutional, behavioural, and temporal—the framework gave me the vocabulary to name what I previously only intuited.”

**\*\*Table 2. Three-Track Output Distribution (2023 – 2025)\*\***

| Track | Output Type            | Proportion | Representative Product                                  |
|-------|------------------------|------------|---------------------------------------------------------|
| A     | Research paper         | ~40%       | SWM problem analysis using diagnostic matrix            |
| B     | Engineering design     | ~35%       | Facility process design with 3D framework justification |
| C     | Pedagogical reflection | ~25%       | Meta-analysis of personal learning trajectory           |

## Discussion

### A Cognitive Progression, Not an Activity Collection

The NFSC pedagogy’s core contribution is not any single teaching technique—narrative excerpts, analytical frameworks, scenario

simulations, and creative outputs all exist independently in the pedagogical literature—but rather the systematic *\*sequencing\** of these elements into a coherent cognitive progression. Narrative triggers affective engagement (Bruner, 1986); framework provides cognitive structure (Perkins, 1991); scenario enables situated practice (Lave & Wenger, 1991); creation closes the loop through productive output (Spady, 1994; Biggs & Tang, 2011). The innovation lies in the *\*order\**: motivation before tools, tools before practice, practice before production.

This ordering directly addresses a limitation shared by many existing pedagogical approaches. PBL and case-based teaching, while effective, typically assume that the *\*motivation to analyse\** is already present—either intrinsically or through the assessment structure (Hmelo-Silver, 2004). NFSC explicitly builds this motivation in Week 1 through narrative engagement, ensuring that when analytical frameworks arrive in Week 2, students are primed to receive them not as abstract tools but as keys to understanding stories they already care about.

### ### Transferability and Boundary Conditions

The NFSC pedagogy's transferability to other engineering disciplines depends on three preconditions. First, the course must possess or develop a "narrativisable" resource base—authentic, emotionally textured accounts of professional practice. This need not involve published documentary novels; practitioner interviews, case archives, or student-collected narratives could serve the same function. Second, the discipline's core knowledge must be abstractable into one or more diagnostic frameworks—structured analytical tools that students can apply across multiple contexts. Third, instructors must be capable of designing and facilitating scenario-based teaching, which represents the highest-demand component of the NFSC model.

A significant boundary condition is the resource intensity of the narrative component. The seven documentary novels used in this study

required over a decade of writing and publication—a commitment unlikely to be replicated by other instructors (Huang, 2025a). However, the underlying \*principle\*—using authentic narrative to establish affective engagement before introducing analytical tools—is resource-independent. Short practitioner narratives, student-collected case stories, or publicly available documentary media could serve the same pedagogical function at lower cost.

### ### Limitations and Reflexivity

This study carries significant limitations that must be transparently stated. The single-course, single-institution design with small cohorts (~30 per cycle) precludes statistical generalisation. The absence of a controlled comparison means that observed cognitive changes cannot be causally attributed to the NFSC pedagogy—novelty effects, instructor personality, cohort composition, and concurrent coursework may all contribute. The instructor’s dual role as teacher and researcher introduces potential confirmation bias, which we have attempted to mitigate through independent coding and member checking but cannot eliminate.

These limitations position the study as exploratory practitioner research rather than confirmatory scholarship (Cochran-Smith & Lytle, 2009). Its value lies in the thick description of a pedagogical innovation and its apparent effects in authentic classroom contexts—documentation that may inform other practitioners’ reflection and adaptation, and that may motivate larger-scale investigation with more rigorous designs.

### ## Conclusion

The NFSC pedagogy offers a theoretically grounded, practically implemented teaching model that addresses the structural gap between knowledge transmission and problem-solving capability in graduate

engineering classrooms. Its distinctive contribution is the articulation of a cognitive progression—narrative → framework → scenario → creation—that systematically builds motivation, analytical tools, applied practice, and creative production in sequence.

The model's transparent acknowledgment of its limitations—small scale, absence of controls, resource-intensity barriers—positions it as an invitation rather than a prescription. Other engineering educators are encouraged to adapt the NFSC design principles to their own disciplinary contexts. The specific implementation—narrative resources, analytical frameworks, scenario types, output tracks—must be reparameterised for each new domain, but the underlying progression logic is transferable.

Future research should pursue quasi-experimental designs comparing NFSC with traditional lecture-based instruction in comparable courses, ideally across multiple institutions. Longitudinal tracking of graduates could assess whether framework-based diagnostic thinking persists beyond the course context into professional practice. Cross-disciplinary adaptations in hydraulic engineering, transportation engineering, and environmental management offer natural extensions for testing the model's generalisability.

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## Declarations

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**\*\*Data Availability Statement.\*\*** The qualitative data supporting this study (student outputs, observation notes) are not publicly available due to privacy considerations. Anonymised excerpts are available from the corresponding author upon reasonable request.

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